Math 3
9-2 Using Patterns to Write and Evaluate Expressions

Name:_____
Date: _____ Per: ___

For problems 1 - 6, use the figures below.

Square Numbers:



1) How many dots are in the 4th, 5th, and 6th figures?

2) Does this pattern have a constant difference? Explain!

3) Describe the pattern in words.

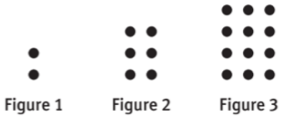
4) How many dots would be in the 10th figure?

5) How many dots are in the 40th figure?

6) How many dots are in the nth figure?

For problems 7 - 9 use the figures below

Rectangular Numbers:



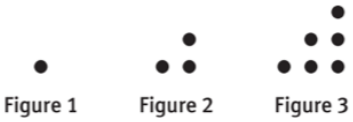
7) Does this pattern have a constant difference? Explain!

8) Make a conjecture about how many dots would be in the nth figure.

9) How many dots are in the 99th figure?

For problems 10 and 11, use the figures below.

Triangular Numbers:



10) How is this pattern related to the one at the top of this page?

11) Write an expression to determine how many dots would be in the nth figure.

Turn Over

Practice:

1) Using the formulas from your notes, determine whether 72 is a square number, rectangular number, or triangular number. Show work.

SQ: $n^2 = 72$ ... no

RECT $n(n+1) = 72$, $8(9) = 72$ ✓

TRI: $\frac{n(n-1)}{2} = 72$, $n(n+1) = 144$ ... no,

For problems 2 - 6 , use the figures below. Assume each side of the pentagon is 1 cm.

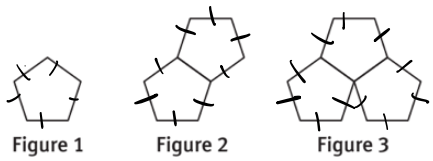


Figure 1 Figure 2 Figure 3

2) What is the perimeter of each figure?

1: 5 cm 2: 8 cm 3: 11 cm

PATTERN: ADD 3 EACH TIME

3) Determine the perimeter of figures 4 - 6.

4: 14 cm 5: 17 cm 6: 20 cm

4) Organize your results from items 2 and 3 into the table below. What would be the perimeter of the 10th figure? Explain.

Figure Number (n)	Perimeter	Difference in Perimeter
1	5	———
2	8	3
3	11	3
4	14	3
5	17	3
6	20	3
7	23	
8	26	
9	29	
10	32	

CONTINUOUSLY ADD 3 UNTIL @ 10th FIGURE

10: 32 cm

5) Use 'n' to represent the figure number. Write an expression that could be used to determine the perimeter of the nth figure.

$5+3(n-1)$ OR $3n+2$

6) Use your expression to determine the perimeter of the 50th figure.

$3(50)+2$

$150+2$

152 cm

For problems 7 -10 , use the figures below. Assume each octagon has a length of 1 m.

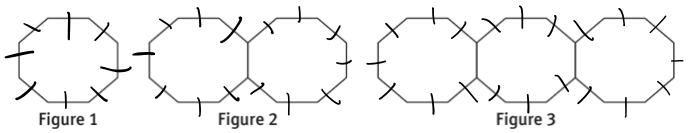


Figure 1 Figure 2 Figure 3

7) What is the perimeter of each figure?

1: 8 m 2: 14 m 3: 20 m

(ADD 6 EACH TIME)

8) Determine the perimeter of figures 4 - 6.

4: 26 m

5: 32 m

6: 38 m

9) Let 'n' represent the figure number. Write an expression that could be used to determine the perimeter of the nth figure.

$8+6(n-1)$ OR $6n+2$

10) Find the perimeter of a walkway made up of 40 of these octagonal blocks.

$6(40)+2$

$240+2$

242 m